Kanna Ruthwik

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.96
Sri Chaitanya College of Education	Class XII (Telangana Board of Intermediate Education)	2023	97.00%
Delhi Public School Warangal	Class X (Central Board of Secondary Education)	2021	96.40%

INTERNSHIP EXPERIENCE

Technical Intern -- Siemens Healthineers

- Developed and adapted **EDM and Consistency Model-based diffusion frameworks** for one-step MRI deblurring, replacing iterative noise-based sampling with a blur-degradation trajectory for accelerated image restoration.
- Designed **Karras -sampling and perturbed- training strategies** to improve restoration under unknown blur levels, and built a **super-resolution pipeline** combining bicubic upscaling with neural deblurring to recover high-frequency anatomical details.
- Developed an **LLM-based MRI motion-artifact scoring system** using zero-shot, few-shot, and contextual prompting; built an interactive **Streamlit dashboard** for visualizing MR images and automated motion scores alongside radiologist assessments.

PROJECTS

Veduify -- AI-Powered Self-Study Platform

- Developed an AI-powered self-study platform using **Next.js, React, and TypeScript**, featuring interactive dashboards and personalized learning workflows.
- Integrated **Google Gemini and Firebase** to enable AI-driven course generation and learning management.

StethoWireAI -- Android Medical Report Summarizer

- Developed an AI-powered Android application using the **Gemini API** to extract and summarize insights from medical reports, with a patient reminder system for adherence.
- Implemented a **Room DAO-based local data architecture** supporting secure multi-member data isolation and scalable data management.

OpsinVision -- Medical Image Processing Android App

- Developed an **Android application in Java and XML** for medical image processing, providing an interactive interface for working with medical images.
- Implemented the **mobile application workflow and image-processing functionality**, integrating image analysis into an accessible Android-based platform.

AI Knowledge Assistant -- RAG

- Built a **Retrieval-Augmented Generation (RAG)** assistant for querying PDF documents using **LangChain, ChromaDB, and Gemini**.
- Implemented **2D/3D vector visualization** to explore document embeddings and retrieval behavior.

Wikipedia Search Application

- Developed a responsive web application integrating the **Wikipedia API** to asynchronously fetch and display search results using JavaScript.
- Implemented dynamic result rendering, responsive UI with **HTML, CSS and Bootstrap**, and direct navigation to

relevant articles.

POSITIONS OF RESPONSIBILITY

- **Class Representative**, Biomedical Engineering -- IIT Hyderabad
- **Web Domain Head**, Ek Bharat Shreshtha Bharat (IIT Hyderabad)
- **Web Coordinator**, Ek Bharat Shreshtha Bharat (IIT Hyderabad)

SKILLS

- **Languages:** Python, C++, Java, JavaScript, SQL, HTML, CSS, React.js, Node.js, Express.js, REST APIs
- **AI & Machine Learning:** PyTorch, TensorFlow, Scikit-learn, Hugging Face, LangChain, Pandas, NumPy, Matplotlib
- **Databases & Cloud:** MySQL, SQLite, MariaDB, Firebase, AWS, Docker, Kubernetes, Linux, Git
- **Automation:** n8n, Make.com

SCHOLASTIC ACHIEVEMENTS

- Selected for IIT Hyderabad's **BUILD Cohort 2024** and **awarded 10,000** funding for **Veduify**, an AI-powered self-study platform.

RELEVANT COURSES

- **Computer Science & AI:** Data Structures & Applications, Algorithms, Database Management Systems, Artificial Intelligence, Probability & Random Processes, Data Visualization
- **Biomedical Engineering:** Bioinformatics, Bioimaging, Sensors & Transducers in Healthcare, Biomechanics, Biomaterials, Digital Fabrication

EXTRACURRICULAR

- **Veghan Naturals and Beverages Pvt. Ltd.** -- Contributed to product branding and design, including label design, packaging, and product strategy.
- **Hackathons & Student Initiatives** -- Participated in technical and student-led initiatives, applying software development and API integration skills to practical problems.
- **Creative & Cross-functional Activities** -- Collaborated across technical, creative, and entrepreneurial initiatives at IIT Hyderabad.